

CoP SAP 6.5 and 6.6 Appendix

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6.5 Appendix

As an appendix to section 6.5, we repeat the analysis using binned dose scores. The single-dose halved dose scores are binned as follows: 0, 1, (1, 1.5], (1.5, 2], (2, 2.5], (2.5, 3], > 3. This is being done in an attempt to improve the performance of the bootstrap confidence intervals.

Numeric viral load

There is one pair for which the best fit model is a 4-parameter logistic. Since it is not straightforward to calculate the percent variance explain for that model (since it is not clear how to make the fuller model that adjusts for covariates with that model), we instead use the second-best model for that pair.

Table A6.5.1: Percent of Variance Explained by Immune Virus-Specific Antibodies - Numeric Viral Load with Binned Dose Score

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
AUC BAL N1	89.94 (77.43, 100)	97.62 (93.99, 100)	89.83 (77.75, 100)	92.46 (85.34, 100)	79.84 (65.78, 100)	78.31 (63.61, 100)	80.92 (66.7, 100)	87.16 (77.03, 100)
AUC BAL Subgenomic	89.47 (79.89, 100)	100 (96.47, 100)	91.09 (79.56, 100)	94.19 (85.36, 100)	70.4 (52.38, 100)	64.13 (43.08, 100)	76.43 (58.63, 100)	82.07 (66.02, 100)
AUC Nasal Swab N1	93.81 (87.69, 100)	94.69 (90.61, 100)	93.42 (89.02, 100)	90.28 (84.41, 100)	85.79 (77.38, 99.99)	92.14 (85.11, 100)	93.51 (89.03, 100)	99.61 (97.13, 100)
AUC Nasal Swab Subgenomic	100 (96.63, 100)	95.68 (92.04, 100)	96.89 (92.28, 100)	93.71 (87.46, 100)	95.06 (90.09, 100)	94.32 (87.25, 100)	100 (97.35, 100)	100 (95.62, 100)
AUC OP Swab N1	99.3 (95.41, 100)	96.69 (92.59, 100)	97.34 (90.71, 100)	95.39 (88.27, 100)	93.31 (85.25, 100)	77.85 (37.29, 100)	94.52 (87.45, 100)	98.17 (93.01, 100)
AUC OP Swab Subgenomic	100 (88.58, 100)	95.08 (87.76, 100)	98.45 (90.69, 100)	97.52 (89.6, 100)	95.99 (86.72, 100)	66.43 (14.32, 100)	99.83 (92.56, 100)	100 (91.3, 100)

log VL Day2	87.48	100	73.4	74.92	95.72	85.99	90.89	93.44
PostChal-	(73.89,	(96.73,	(56.17,	(57.45,	(88.42,	(73.96,	(80.73,	(85.01,
lenge BAL	100)	100)	100)	100)	100)	100)	100)	100)
N1								
log VL Day2	94.95	100	75.46	77.27	95.23	90.61	93.43	95.63
PostChal-	(87.67,	(95.92,	(57.94,	(60.91,	(85.27,	(77.49,	(82.82,	(87.29,
lenge BAL	100)	100)	100)	100)	100)	100)	100)	100)
Subgenomic								
log VL Day2	86.82	81.86	86.61	75.55	100	97.27	100	100
PostChal-	(77.97,	(65.35,	(68.78,	(55.82,	(87.22,	(88.94,	(83.5,	(88.15,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
OPswab N1								
log VL Day2	91.53	91.35	88.21	79	100	100	100	100
PostChal-	(80.42,	(80.38,	(72.25,	(58.44,	(89.38,	(87.31,	(85.44,	(86.27,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
OPswab								
Subgenomic								
log VL Day2	99.75	95.94	93.41	88.64	100	96.21	100	100
PostChal-	(96.21,	(92.55,	(87.4,	(82.13,	(98.08,	(89.34,	(93.86,	(97.72,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
NasalSwab								
N1								
log VL Day2	100	99.9	95.69	90.33	100	97.29	100	100
PostChal-	(96.37,	(98.62,	(92,	(86.23,	(96.83,	(92.16,	(93.36,	(95.81,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
NasalSwab								
Subgenomic								

Binary Viral Load

Table A6.5.2: Percent of Variance Explained by Immune Virus Specific Antibodies - Binary Viral Load with Binned Dose Score

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
bin VL Day2	100	70.74	57.51	36.28	100	100	100	94.71
PostChal-	(48.01,	(33.67,	(14.04,	(5.42,	(58.92,	(24.01,	(47.68,	(61.42,
lenge BAL	100)	100)	100)	100)	100)	100)	100)	100)
N1								
bin VL Day2	96.86	100	63.4	59.58	100	89.43	90.74	97.98
PostChal-	(94.09,	(93.55,	(45.07,	(39.98,	(94.94,	(81.27,	(82.7,	(93.07,
lenge BAL	100)	100)	100)	100)	100)	100)	100)	100)
Subgenomic								
bin VL Day2	100	100	100	100	92.26	100	100	100
PostChal-	(87.62,	(61.4,	(57.39,	(55.04,	(57.98,	(68.57,	(52.99,	(58.62,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
OPswab N1								

bin VL Day2	56.34	45.05	35.52	26.85	73.99	85.89	69.65	69.65
PostChal- lenge	(31.92, 100)	(18.65, 100)	(6.2, 100)	(0, 77.48)	(56.01, 100)	(75.35, 100)	(50.74, 100)	(50.28, 100)
OPswab								
Subgenomic								
bin VL Day2	6.4 (0,	17.42	80.9	0 (0,	9.83	1.23	6.47	9.37
PostChal- lenge	100)	(4.34, 53.78)	(0, 100)	100)	(0, 100)	(0, 100)	(0, 100)	(0, 100)
NasalSwab								
N1								
bin VL Day2	100	100	100	100	100	100	100	100
PostChal- lenge	(81.43, 100)	(71.17, 100)	(71.08, 100)	(59.89, 100)	(85.35, 100)	(83.58, 100)	(64.4, 100)	(77.26, 100)
NasalSwab								
Subgenomic								

6.6 Appendix

SAP section 6.6 is also repeated using larger bins of the single dose halved dose scores: 0, 1, (1, 2], >2. This was done to increase the amount of data in each bin, after splitting by vaccine.

Numeric Viral Load

Table A6.6.1: Differential Vaccine Effect - Numeric Viral Load with Binned Dose Score

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
AUC BAL N1	90.83 (81.78, 100)	100 (98.74, 100)	82.2 (71.22, 98.49)	91.12 (86.59, 100)	90.05 (83.97, 100)	94.08 (90.6, 100)	87.61 (81.78, 100)	99.17 (97.17, 100)
AUC BAL Subgenomic	98.51 (90.88, 100)	100 (92.39, 100)	89.43 (81.26, 100)	99 (96.66, 100)	100 (94.12, 100)	100 (91.02, 100)	92.96 (85.66, 100)	100 (88.47, 100)
AUC Nasal Swab N1	73.23 (59.49, 88.79)	74.78 (63.87, 90.93)	74.87 (61.2, 89.12)	82.1 (73.1, 94.6)	76.63 (64.24, 91.77)	77.08 (65.82, 92.99)	74.05 (61.4, 88.19)	79.03 (67.49, 92.75)
AUC Nasal Swab	86.82 (79.81, 99.26)	88.36 (82.25, 100)	81.52 (71.94, 96.26)	87.33 (80.52, 100)	83.07 (74.45, 97.48)	82.26 (71.21, 96.1)	81.41 (72.29, 96.06)	89.08 (83.46, 100)
AUC OP Swab N1	67.88 (41.02, 92)	71.5 (41.39, 94.19)	62.24 (36.79, 84.79)	63.52 (36.06, 86.59)	65.89 (38.49, 91.26)	66.53 (37.82, 91.12)	62.73 (35.27, 87.41)	71.23 (42.62, 94.61)
AUC OP Swab	51.83 (23.64, 87.28)	56.75 (25.5, 90.21)	43.9 (19, 76)	46.46 (22.26, 82.3)	46.93 (22.19, 86.66)	45.59 (13.27, 75.82)	43.66 (17.72, 80.29)	51.15 (21.64, 88.04)

log VL Day2	91.18	90.76	84.13	91.05	95.2	88.54	93.02	95.63
PostChal-	(82.72,	(86.8,	(75.23,	(87.88,	(91.94,	(82.92,	(88.59,	(91.83,
lenge BAL	100)	100)	100)	100)	100)	100)	100)	100)
N1								
log VL Day2	100	100	93.02	97.3	100	100	100	100
PostChal-	(95.7,	(96.58,	(90.62,	(95.77,	(94.18,	(93.65,	(99.31,	(96.05,
lenge BAL	100)	100)	100)	100)	100)	100)	100)	100)
Subgenomic								
log VL Day2	62.63	48.73	58.13	57.89	64.07	48.9	66.93	74.42
PostChal-	(44.15,	(24.74,	(40.65,	(37.44,	(43.9,	(26.66,	(49.33,	(60.09,
lenge	100)	84.16)	100)	90.53)	100)	81.38)	100)	100)
OPswab N1								
log VL Day2	75.54	65.74	67.52	65.49	78.98	61.01	75.14	86.95
PostChal-	(62.13,	(48.9,	(51.14,	(46.92,	(67.89,	(40.18,	(63.65,	(78.79,
lenge	100)	100)	100)	96.61)	100)	92.85)	100)	100)
OPswab								
Subgenomic								
log VL Day2	100	85.43	97.27	100	100	90.2	100	100
PostChal-	(94.25,	(77.89,	(94.49,	(95,	(92.94,	(83.01,	(95.48,	(93.38,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
NasalSwab								
N1								
log VL Day2	99.79	85.99	96.38	98.43	100	86.83	100	100
PostChal-	(96.45,	(81.05,	(94.04,	(95.07,	(97.69,	(79.01,	(96.66,	(97.14,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
NasalSwab								
Subgenomic								

Binary Viral Load

Table A6.6.2: Differential Vaccine Effect - Binary Viral Load with Binned Dose Score

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
bin VL Day2	0 (0,	0 (0,	100 (0,	90.74	100 (0,	0 (0,	0 (0,	100 (0,
PostChal-	100)	100)	100)	(0,	100)	100)	100)	100)
lenge BAL				100)				
N1								
bin VL Day2	100	93.71	100	100	97.38	94.91	97.26	67.61
PostChal-	(83.85,	(75.41,	(86.32,	(85.07,	(77.61,	(72.23,	(80.34,	(36.24,
lenge BAL	100)	100)	100)	100)	100)	100)	100)	100)
Subgenomic								
bin VL Day2	0 (0,	57.04	0 (0,	19.65	35.89	0 (0,	26.05	45.58
PostChal-	100)	(0,	100)	(0,	(2.07,	100)	(0,	(0,
lenge		100)		91.66)	100)		100)	100)
OPswab N1								

bin VL Day2	74.61	56.8	72.56	84.63	73.26	55.23	58.52	95
PostChal- lenge	(54.02, 100)	(27.77, 100)	(53.89, 100)	(62.91, 100)	(51.04, 100)	(28.63, 100)	(29.98, 100)	(75.25, 100)
OPswab								
Subgenomic								
bin VL Day2	11.36	100	0 (0,	0 (0,	16.01	0 (0,	17.23	6.45
PostChal- lenge	(0, 100)	(100, 100)	100)	100)	(0, 100)	100)	(0, 100)	(0, 100)
NasalSwab								
N1								
bin VL Day2	0 (0,	24.25	100	0 (0,	65.77	46.62	100	66.97
PostChal- lenge	100)	(0, 53.89)	(58.79, 100)	100)	(25.11, 100)	(14.1, 100)	(52.29, 100)	(26.99, 100)
NasalSwab								
Subgenomic								

Additional Analysis - Base+Dose+Vacc vs. Base

As an additional analysis for the appendix, we compare the base model to the base+dose+vacc model. Details of the base model are explained in the SAP section 6.5. The base+dose+vacc model used here is slightly different than that described in SAP section 6.6. In this model, the single dose halved dose score is used as a numeric value, with a separate effect for each vaccine manufacturer. Linear models for numeric viral load (e.g., Section 6.4.1) or logistic models for binary viral load (more complicated models, regardless of what was selected by the cross-validation procedure in section 6.4.2, are not used) will be employed. The percentage of variance (PVE) is 100 times the adjusted R square value from the base model over the adjusted R squared values from the base+dose+vacc model. Confidence intervals on the PVE are calculated by using bias-adjusted nonparametric bootstrap methods (using 5000 replicates).

Numeric Viral Load

Table A6.6.3: Compare Base to Base, Dose, Vacc - Numeric Viral Load with Binned Dose Score

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
AUC BAL	88.8	98.2	77.71	85.42	71.53	71.43	74.57	85.9
N1	(81.23, 100)	(96.63, 100)	(62.88, 96.09)	(76.65, 100)	(62.09, 94.64)	(60.72, 98.51)	(65.04, 100)	(79.74, 100)
AUC BAL	91.92	97.25	80.09	89.19	66.8	63.02	71.54	83.68
Subgenomic	(85.6, 100)	(93.52, 100)	(63.3, 100)	(77.58, 100)	(56.09, 100)	(49.37, 100)	(60.38, 100)	(73.51, 100)
AUC Nasal	77.44	78.51	72.73	74.44	74.32	74.52	76.64	84.57
Swab N1	(67.5, 91.85)	(69.86, 95.34)	(62.82, 86.61)	(65.03, 88.82)	(64.72, 90.73)	(65.38, 91.74)	(67.79, 91.24)	(76.9, 98.19)
AUC Nasal	88.38	85.29	73.88	76.66	79.74	77.34	81.68	90.64
Swab	(82.98, 99.07)	(79.61, 99.33)	(62.91, 90.36)	(66.94, 91.73)	(72.71, 94.36)	(67.72, 93.07)	(74.48, 92.93)	(86.61, 99.56)
Subgenomic								

AUC OP Swab N1	82.53 (72.82, 100)	92.28 (86.48, 100)	76.81 (62.11, 94.82)	83.08 (71.26, 100)	76.46 (69.89, 95.33)	71.1 (54.25, 94.11)	83.63 (79, 100)	90.88 (86.33, 100)
AUC OP Swab Subgenomic	82.55 (70.55, 100)	77.24 (44.5, 100)	71.52 (47.11, 94.7)	79.33 (58.84, 100)	70.62 (59.3, 98.01)	50.29 (13.47, 80.26)	78.56 (69.63, 100)	82.46 (73.29, 100)
log VL Day2 PostChal-enge BAL N1	83.29 (75.83, 100)	98.13 (97.55, 100)	65.1 (51.39, 91.13)	73.82 (65.47, 97.18)	90.1 (87.76, 100)	84.67 (82, 100)	88.23 (84.16, 100)	92.5 (91.65, 100)
log VL Day2 PostChal-enge BAL Subgenomic	86.67 (80.05, 100)	100 (98.36, 100)	64.54 (48.56, 91.14)	70.41 (56.83, 100)	88.66 (82.64, 100)	87.88 (82.15, 100)	89.44 (82.98, 100)	89.96 (83.45, 100)
log VL Day2 PostChal-enge OPswab N1	74.66 (65.32, 100)	69.86 (59.88, 100)	60.18 (43.55, 100)	51.88 (33.83, 100)	76.82 (65.84, 100)	74.18 (62.21, 100)	82.62 (75.78, 100)	92.8 (85.98, 100)
log VL Day2 PostChal-enge OPswab Subgenomic	77.99 (66.4, 100)	78.61 (71.45, 100)	68.61 (51.21, 100)	59.14 (39.53, 100)	88.5 (84.59, 100)	84.97 (80.16, 100)	88.99 (84.86, 100)	100 (97.59, 100)
log VL Day2 PostChal-enge NasalSwab N1	99.78 (94.37, 100)	100 (98.13, 100)	81.02 (73.62, 100)	78.98 (69.81, 100)	100 (98.32, 100)	94.9 (90.54, 100)	100 (95.88, 100)	100 (93.21, 100)
log VL Day2 PostChal-enge NasalSwab Subgenomic	97.53 (93.73, 100)	99.21 (97.65, 100)	81.3 (76.45, 100)	78.89 (72.7, 100)	96.29 (94.4, 100)	93.08 (88.54, 100)	99.74 (99.11, 100)	100 (96.06, 100)

Binary Viral Load

Table A6.6.4: Compare Base to Base, Dose, Vacc - Binary Viral Load with Binned Dose Score

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
bin VL Day2 PostChal-enge BAL N1	100 (41.59, 100)	100 (58.73, 100)	47.52 (19.38, 100)	39.4 (21.55, 100)	100 (54.97, 100)	100 (40.25, 100)	100 (31.37, 100)	94.41 (52.57, 100)

bin VL Day2	78.31	100	56.34	56.84	91.82	95.69	80.1	90.63
PostChal-	(72.4,	(93.45,	(42.3,	(42.28,	(91.8,	(99.49,	(75.17,	(90.36,
lenge BAL	100)	100)	95.48)	93.22)	100)	100)	100)	100)
Subgenomic								
bin VL Day2	67.02	100	47.13	48.92	46.6	63.65	65.52	75.53
PostChal-	(36.29,	(66.55,	(14.35,	(18.68,	(19.5,	(28.73,	(32.39,	(43.42,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
OPswab N1								
bin VL Day2	51.17	41.56	28.27	29.51	62.51	66.49	53.74	74.4
PostChal-	(37.48,	(23.17,	(8.42,	(8.48,	(50.58,	(59.7,	(41.59,	(67.56,
lenge	100)	100)	67.3)	72.28)	100)	100)	100)	100)
OPswab								
Subgenomic								
bin VL Day2	0 (0,	17.42	0 (0,	0 (0,	3.85	0 (0,	2.92	0 (0,
PostChal-	100)	(4.5,	100)	100)	(0,	100)	(0,	100)
lenge		58.69)			100)		100)	
NasalSwab								
N1								
bin VL Day2	67.83	83.21	58.96	69.04	76.23	58.64	100	78.05
PostChal-	(60.04,	(81.83,	(47.53,	(60.27,	(64.05,	(45.13,	(98.45,	(70.11,
lenge	100)	100)	100)	100)	100)	100)	100)	100)
NasalSwab								
Subgenomic								
